

## AI Quiz 2

1. In supervised learning, “supervision” refers to:

B. Presence of labeled training data

- Slide 2, 3 clearly state: “Training data are labeled” and “accompanied by labels indicating the class of the observations”.

- ☒ A (human monitoring) is not required.
  - ☒ C (iterative updates) is not supervision.
  - ☒ D (normalization) is preprocessing.
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2. Overfitting primarily results from:

C. High variance

- Slide 13: “Variance is the error caused by the model being too sensitive to training data. Result: Overfitting.”

- ☒ A (too simple) → underfitting.
  - ☒ B (high bias) → underfitting.
  - ☒ D (lack of features) → underfitting.
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3. LOOCV is computationally expensive because:

B. It trains N models

- Slide 11: “LOOCV = Train on (N-1) samples, test on 1 sample, repeat N times.”

- ☒ A (large datasets) is not the direct reason; even small N = N models.
- ☒ C (random splits) describes k-fold, not LOOCV.
- ☒ D (ignores training data) is false.

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4. Bias is associated with:

B. Underfitting

- Slide 13: "Bias is the error caused by too simple assumptions. Result: Underfitting."
  -  A (overfitting) → variance.
  -  C (noise learning) → overfitting.
  -  D (data leakage) not related here.
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5. Variance is associated with:

C. Sensitivity to training data

- Slide 13: "Variance is the error caused by the model being too sensitive to training data."
  -  A (simplicity) → bias.
  -  B (underfitting) → bias.
  -  D (feature selection) unrelated.
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6. Distance-weighted k-NN gives:

C. Higher weight to closer neighbors

- Slide 29: "Weight the contribution ... according to their distance ... Give greater weight to closer neighbors."
-  A (equal weight) is standard k-NN, not weighted.
-  B (higher weight to distant) is opposite.
-  D (random) is wrong.

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7. Which metric is best when false positives are costly?

C. Precision

- Slide 33: "Precision tells us how reliable the model's positive predictions are."
  - High precision = low false positives (FP in denominator).
  - ~~✗~~ A (recall) cares about false negatives.
  - ~~✗~~ B (accuracy) hides class issues.
  - ~~✗~~ D (F1) balances both, but precision directly penalizes FP.
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8. Recall measures:

C. Correct positives among actual positives

- Slide 34: "Recall measures how many actual positive cases the model successfully detects."
  - Formula:  $TP / (TP + FN)$ .
  - ~~✗~~ A (correct negatives) is specificity.
  - ~~✗~~ B (correct among predicted positives) is precision.
  - ~~✗~~ D (total accuracy) is accuracy.
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9. Accuracy can be misleading when:

B. Data is imbalanced

- Slide 32-35 example: 95% accuracy but if only 1% were positives, a dummy classifier could score 99% by predicting all NO.
- ~~✗~~ A (normalized) helps, not misleads.

- ☒ C (independent features) is Naïve Bayes assumption.
  - ☒ D (linear model) unrelated.
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10. Which term represents true negatives?

C. Correct negative predictions

- Slide 32: TN = True Negative = actual NO, predicted NO.

- ☒ A (correct positive) = TP.
  - ☒ B (incorrect positive) = FP.
  - ☒ D (incorrect negative) = FN.
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11. Regression is used for:

C. Continuous value prediction

- Slide 37: "Prediction models continuous-valued functions."

- ☒ A (classification) → discrete labels.
  - ☒ B (clustering) → unsupervised.
  - ☒ D (dimensionality reduction) → PCA, etc.
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12. Least squares method minimizes:

B. Squared error

- Slide 38: "Method of least squares estimates the best-fitting straight line" – minimizes sum of squared residuals.

- ☒ A (absolute error) → L1 regression.

- ☒ C (log error) → logistic loss.
  - ☒ D (classification error) → 0/1 loss.
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13. Multiple regression involves:

B. Multiple predictors

- Slide 38: “Multiple linear regression involves more than one predictor variable.”
  - ☒ A (one predictor) → simple linear regression.
  - ☒ C (none) → just mean.
  - ☒ D (only categorical) → ANOVA.
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14. Which step reduces noise?

A. Data cleaning

- Slide 16: “Data cleaning – preprocess data in order to reduce noise and handle missing values.”
  - ☒ B-D are later steps that do not remove noise.
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15. Feature selection helps to:

B. Remove irrelevant features

- Slide 16: “Relevance analysis (feature selection) – Remove irrelevant or redundant attributes.”
- ☒ A (increase noise) opposite.
- ☒ C (increase dimensionality) opposite.
- ☒ D (overfit) – removal reduces overfitting.

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16. Which problem arises due to irrelevant features in k-NN?

B. Curse of dimensionality

- Slide 29: "Curse of dimensionality: distance between neighbors could be dominated by irrelevant attributes."

- ☒ A (overfitting) possible but not primary named issue.
  - ☒ C (underfitting) not typical.
  - ☒ D (bias) not directly.
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17. Normalization is part of:

B. Data transformation

- Slide 16: "Data transformation – Generalize and/or normalize data."

- ☒ A (testing) happens after transformation.
  - ☒ C (model usage) uses transformed data.
  - ☒ D (classification) is the overall task.
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18. Laplace-corrected probability for income=low (k=3)

C. 1/1003

Step-by-step (from Slide 24-26):

- Count of income=low = 0, total instances = 1000, k = 3 possible values (low, medium, high).

- Laplace smoothing formula:

$$P = \text{count} + 1 / \text{total} + k = 0 + 1 / 1000 + 3 = 1/1003$$

-  A (0/1003) is uncorrected.
  -  B (1/1000) would be for  $k=0$ .
  -  D (1/1001) would be for  $k=1$ .
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19. Distinguishes classification from prediction:

B. Classification outputs discrete labels, prediction outputs continuous values

- Slide 4: "Classification predicts categorical class labels. Prediction models continuous-valued functions."

-  A swapped.
  -  C & D both wrong.
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20. Matches bias/variance with training/testing errors:

B. Bias is training errors; Variance is testing errors

- Slide 13: "Bias is Training Errors. Variance is Testing Errors." (exact quote)

-  A reversed.
  -  C both training  $\rightarrow$  wrong.
  -  D both testing  $\rightarrow$  wrong.
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